

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines. Some nodes are highlighted with blue circles, and others with blue dots. The lines are thin and gray, creating a mesh-like structure.

Computer Networking

A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a network of nodes and connections, with several nodes highlighted by blue circles and others by blue dots. The overall style is clean and modern, with a light gray background.

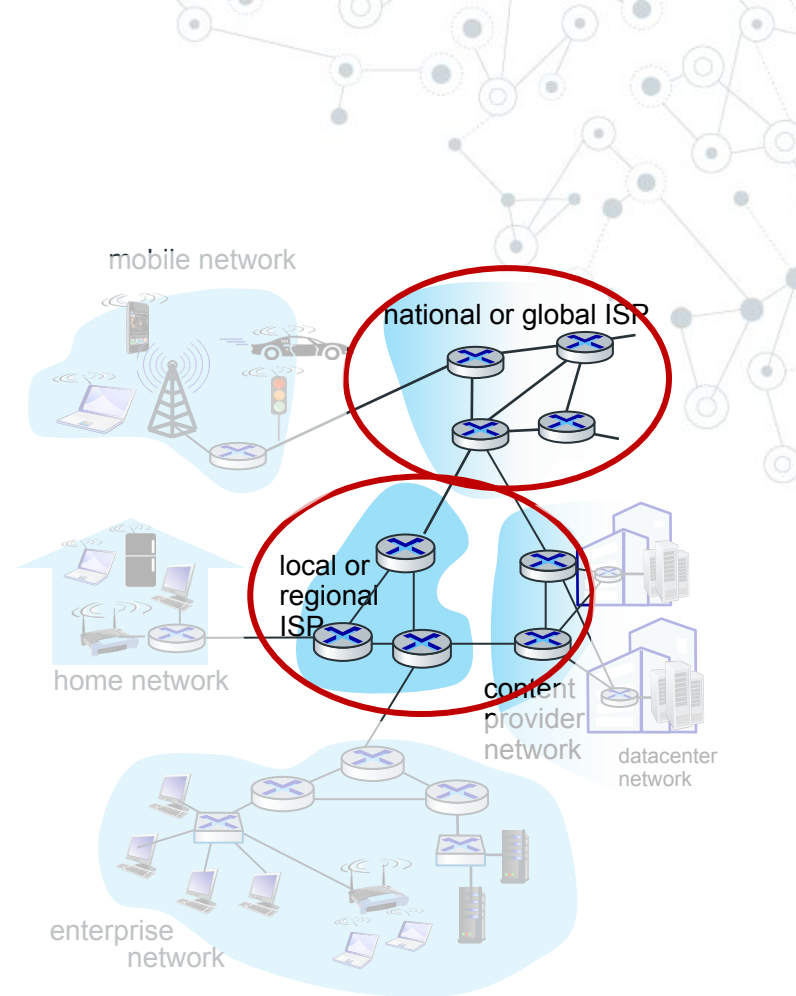
Chapter 1: Computer Networks and the Internet

Overview/roadmap:

- What is the Internet?
- What is a protocol?
- The Network edge: hosts, access network, physical media
- The Network core: packet/circuit switching, internet structure
- Performance: loss, delay, throughput
- Security
- Protocol layers, service models

The network core

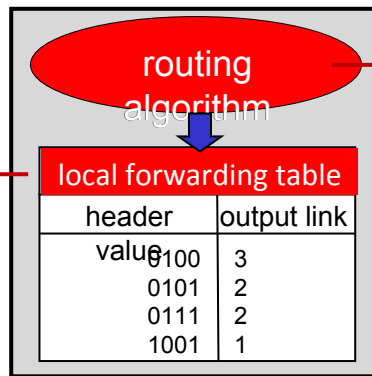
- ◎ mesh of interconnected routers
- ◎ **packet-switching**: hosts break application-layer messages into *packets*
 - forward packets from one router to the next, across links on path from source to destination
 - each packet transmitted at full link capacity



Two key network-core functions

Forwarding:

local action:
move arriving packets from
router's input link to
appropriate
router output link



destination address in arriving
packet's header

Routing:

- **global** action:
determine
source-destination
paths taken by
packets
- routing algorithms

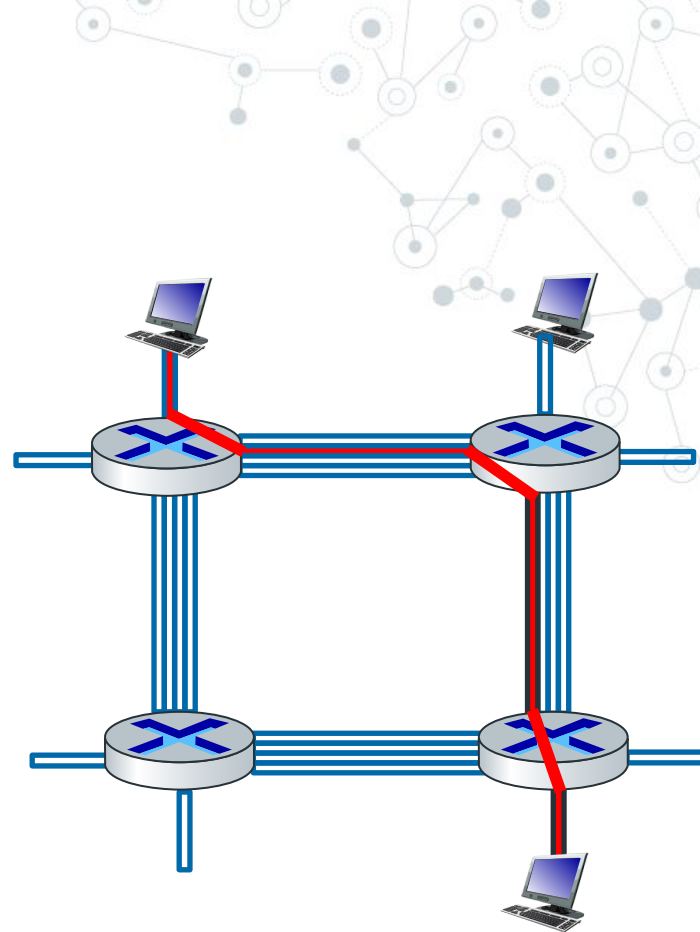
Circuit Switching

- ◎ In circuit-switched networks, the resources needed along a path (buffers, link transmission rate) to provide for communication between the end systems are reserved for the duration of the communication session between the end systems.
- ◎ Example:
 - ◎ ▪ Traditional telephone networks

Circuit Switching

end-end resources allocated to, reserved for “call”
between source and destination

- in diagram, each link has four circuits.
- call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
- circuit-like (guaranteed) performance
- circuit segment idle if not used by call (no sharing)
- commonly used in traditional telephone networks
- if each link between adjacent switches has a



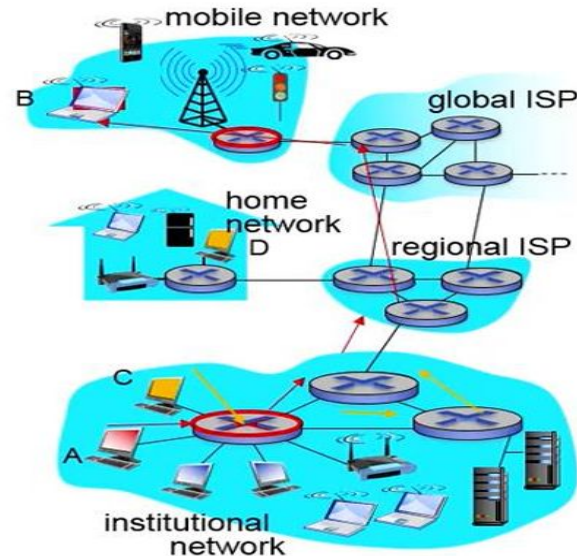
Circuit Switching

circuit switching

end-end resources allocated to, reserved for “call” between source & dest:

- in diagram, each link 2MB
 - A wants to communicate with B, and needs 1.5MB data rate
 - Call setup
 - Transmit data
 - C wants to D, need 1.5MB
 - Callsetup
 - No callsetup
- dedicated resources: no sharing
 - circuit-like (guaranteed) performance

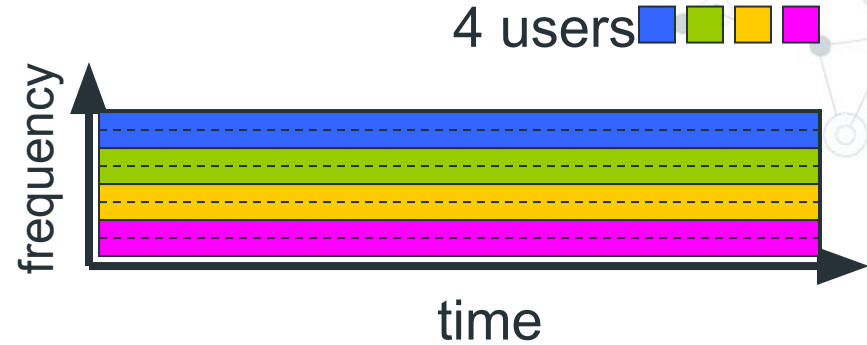
circuit segment idle if not used by call
(no sharing)
commonly used in traditional telephone networks
For continuous nature communication



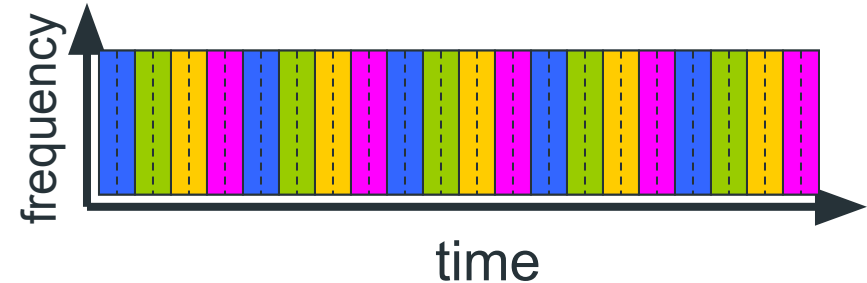
Circuit switching: FDM and TDM

Frequency Division Multiplexing (FDM)

- optical, electromagnetic frequencies divided into (narrow) frequency bands

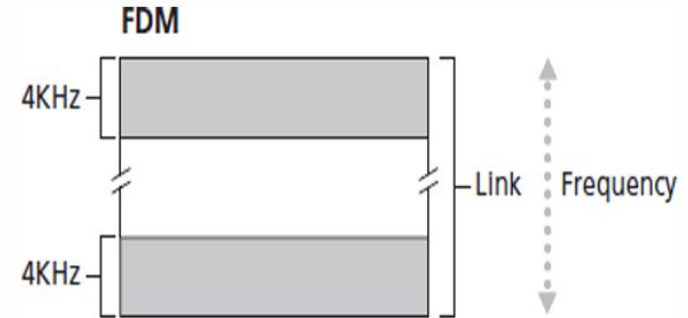


- each call allocated its own band, can transmit at max rate of that narrow band
- Time Division Multiplexing (TDM)
 - time is divided into frames of fixed duration
 - each frame is divided into a fixed number of time slots
 - each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band, but only during its time slot(s)

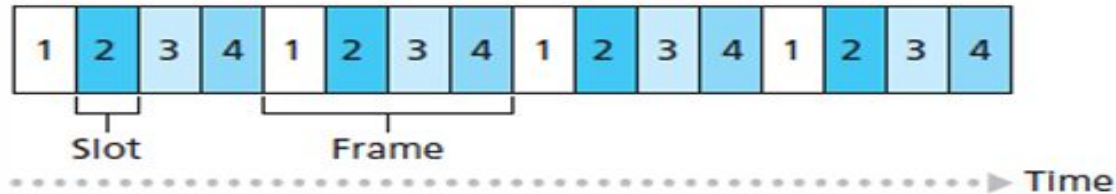


Circuit switching: FDM and TDM

- In telephone networks, this frequency band typically has a width of 4 kHz



TDM



Key:

2 All slots labeled "2" are dedicated to a specific sender-receiver pair.

Circuit switching: FDM and TDM

For TDM, the transmission rate of a circuit is equal to the frame rate multiplied by the number of bits in a slot.

For example, if the link transmits 8,000 frames per second and each slot consists of 8 bits, then the transmission rate of each circuit is 64 kbps.

Circuit switching: FDM and TDM

Example:

Let us consider how long it takes to send a file of 640,000 bits from Host A to Host B over a circuit switched network. Suppose that all links in the network use TDM with 24 slots and have a bit rate of 1.536 Mbps. Also suppose that it takes 500 msec to establish an end-to-end circuit before Host A can begin to transmit the file. How long does it take to send the file?

Solution:

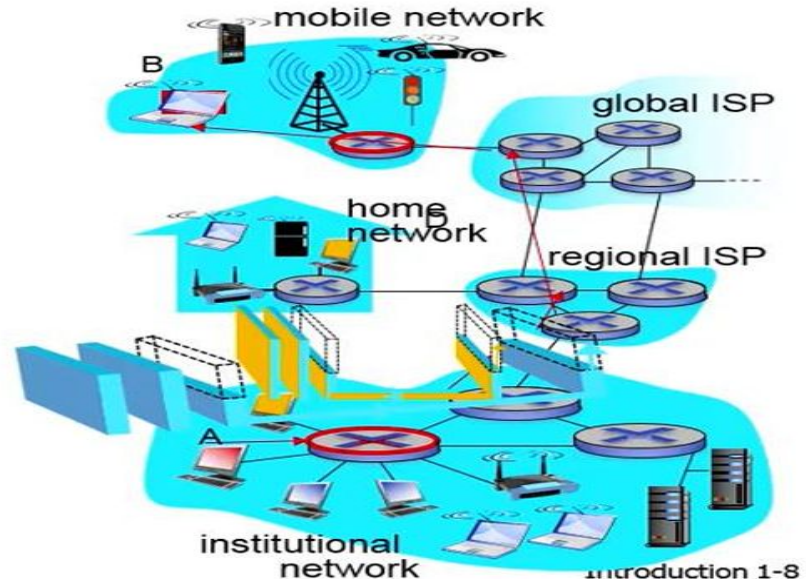
- Each circuit has a transmission rate of:
 $1.536 \text{ Mbps} / 24 = 64 \text{ kbps}$
so it takes $(640,000 \text{ bits}) / (64 \text{ kbps}) = 10 \text{ seconds}$ to transmit the file. To this 10 seconds we add the circuit establishment time, giving 10.5 seconds to send the file.

Packet switching: store and forward

Packet-switching: store-and-forward

- in diagram, each link 2MB
 - A wants to communicate with B, and needs 1.5MB data rate
 - Transmit data
 - C wants to D, need 1.5MB
- Sharing resources
 - No guaranteed delivery

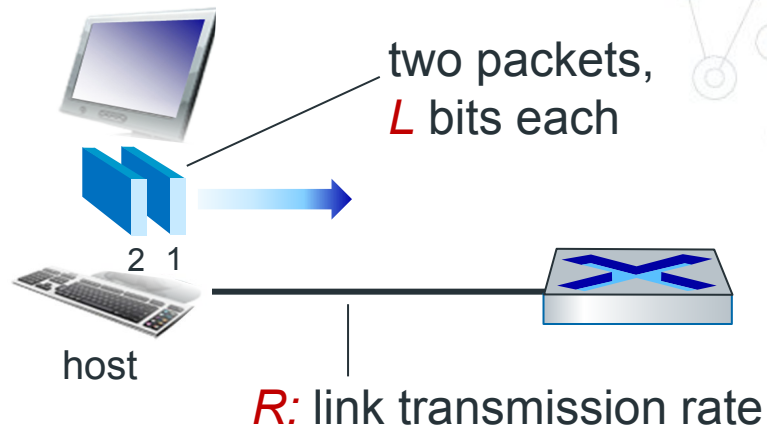
segment idle can be used by others
commonly used in Internet
For bursty data



Host: sends *packets* of data

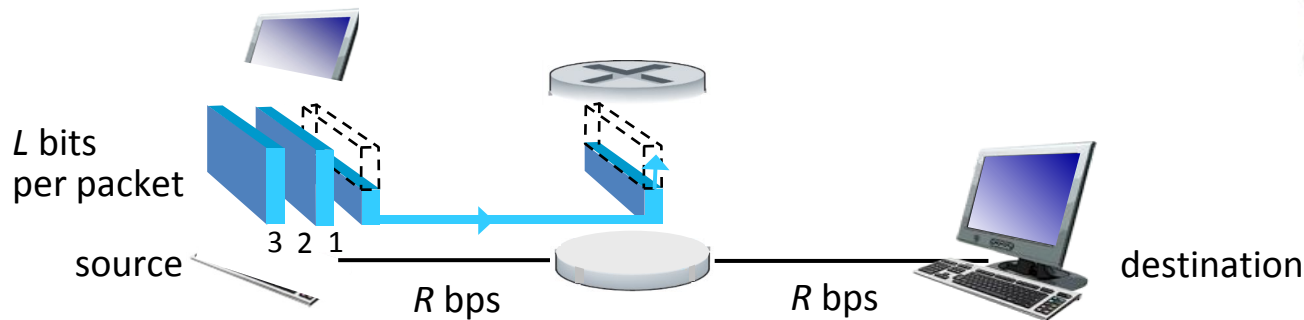
host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length L bits
- transmits packet into access network at *transmission rate R*
 - link transmission rate, aka link *capacity, aka link bandwidth*



$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Packet-switching: store-and-forward



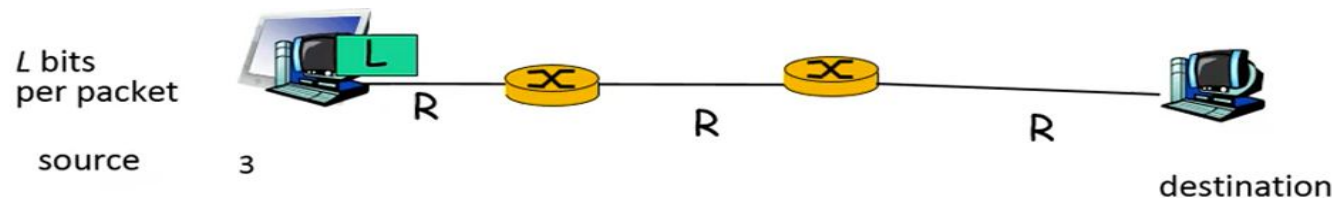
◎ **Transmission delay:** takes L/R seconds to transmit (push out) L -bit packet into link at R bps

◎ **Store and forward:** entire packet must arrive at router before it can be transmitted on next link

One-hop numerical example:

- $L = 10$ Kbits
- $R = 100$ Mbps
- one-hop transmission delay = 0.1 msec

Packet switching: store and forward



- divides into packet
- Each packet has full address
- takes L/R seconds to transmit (push out) L -bit packet into link at R bps
- **store and forward**: entire packet must arrive at router before it can be transmitted on next link
- end-end delay = $3L/R$ (assuming zero propagation delay)

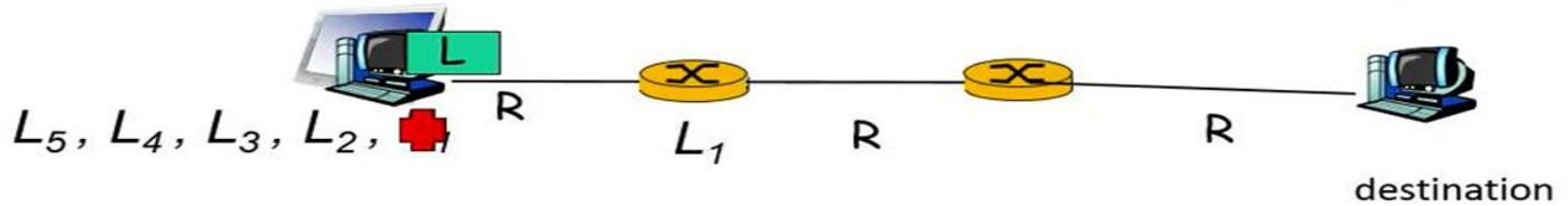
one-hop numerical example:

- $L = 7.5$ Mbits
- $R = 1.5$ Mbps
- one-hop transmission delay = $L/R = 5$ sec
- End-to-end delay = $3L/R = 3 \times 5 = 15$ seconds

} more on delay shortly ...

Introduction 1-9

Packet switching: store and forward



1sec

5 packets example:

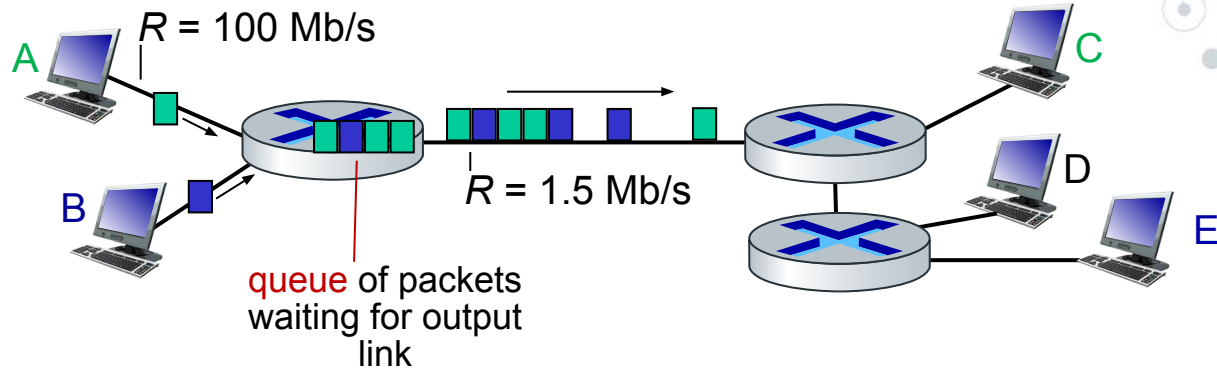
- $L = 7.5$ Mbits
- Divide it in 5 packets, $L_1, L_2, L_3, L_4, L_5, = 7.5/5 = 1.5$ Mbits
- 3 hops delay = 7 sec

One packet example:

- $L = 7.5$ Mbits
- $R = 1.5$ Mbps
- one-hop transmission delay = 5 sec
- 3 hops delay = $3 * 5$

} more on delay shortly ...

Packet-switching: queueing delay, loss



Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for a period of time:

- ⊙ packets will queue, waiting to be transmitted on output link
- ⊙ packets can be dropped (lost) if memory (buffer) in router fills up

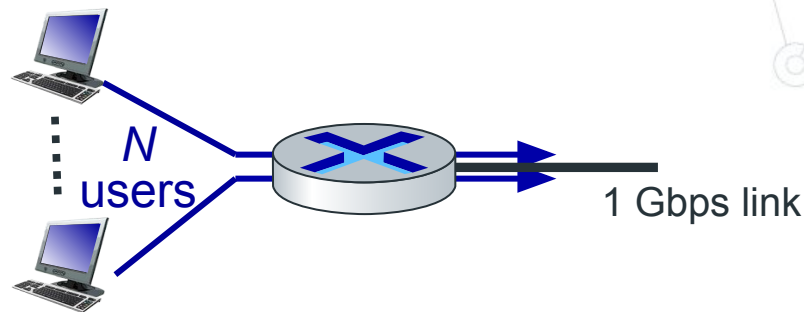
Packet switching versus circuit switching

packet switching allows more users to use network!

Example:

- 1 Gb/s link
- each user:
 - 100 Mb/s when “active”
 - active 10% of time

- *circuit-switching*: 10 users
- *packet switching*: with 35 users, probability > 10 active at same time is less than .0004 *



Q: how did we get value 0.0004?

Q: what happens if > 35 users ?

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive

Packet switching versus circuit switching

Is packet switching a “slam dunk winner”?

- great for “bursty” data – sometimes has data to send, but at other times not
 - resource sharing
 - simpler, no call setup
- **excessive congestion possible:** packet delay and loss due to buffer overflow
 - protocols needed for reliable data transfer, congestion control
- **Q: How to provide circuit-like behavior?**
 - bandwidth guarantees traditionally used for audio/video applications

Q: human analogies of reserved resources (circuit switching)
versus on-demand allocation (packet switching)?

Internet structure: a “network of networks”

- ⊙ Hosts connect to Internet via **access** Internet Service Providers (ISPs)
 - ⊙ residential, enterprise (company, university, commercial) ISPs
- ⊙ Access ISPs in turn must be interconnected
 - ⊙ so that any two hosts can send packets to each other
- ⊙ Resulting network of networks is very complex
 - ⊙ evolution was driven by **economics** and **national policies**
- ⊙ Let's take a stepwise approach to describe current Internet structure

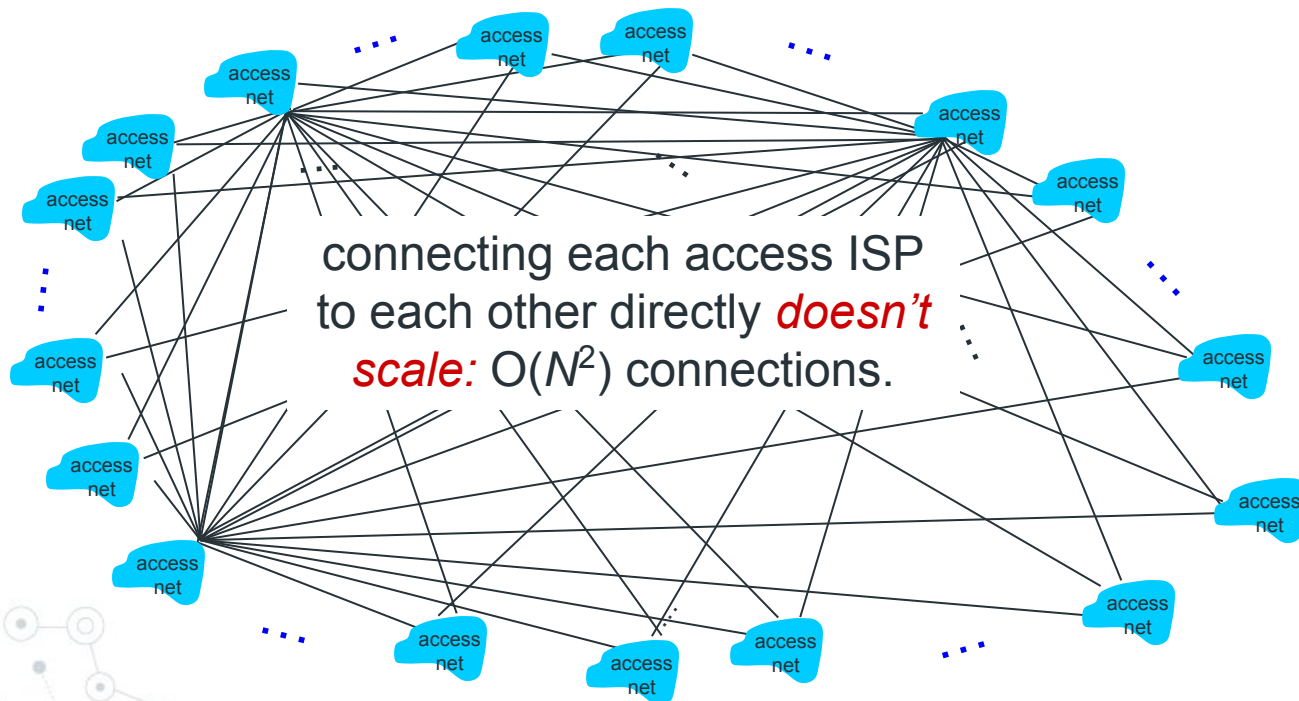
Internet structure: a “network of networks”

Question: given *millions* of access ISPs, how to connect them together?



Internet structure: a “network of networks”

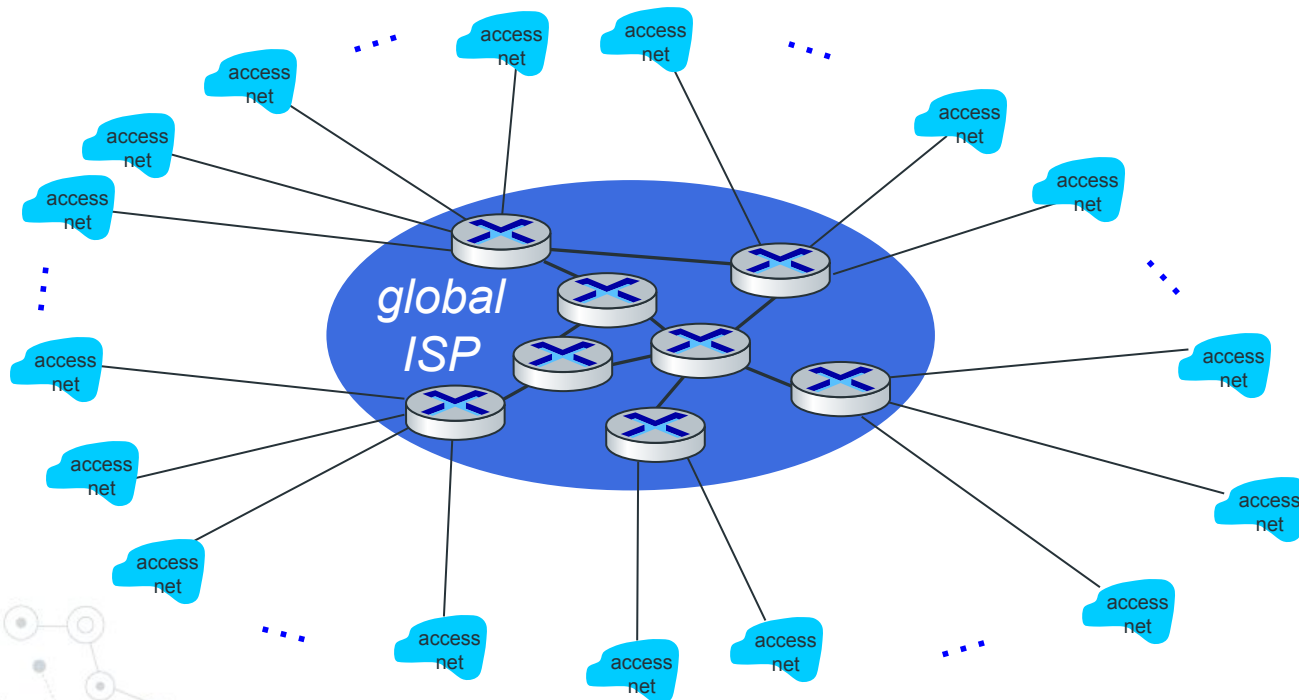
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Internet structure: a “network of networks”

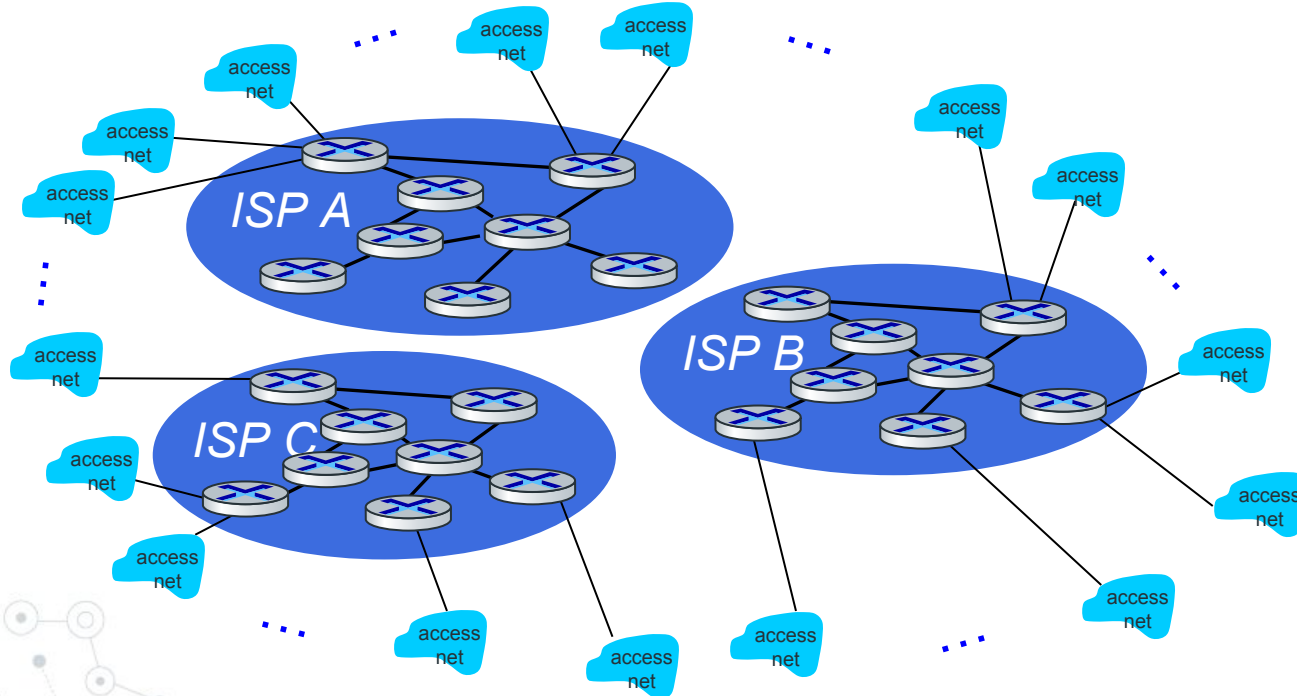
Option: connect each access ISP to one global transit ISP?

Customer and provider ISPs have economic agreement.



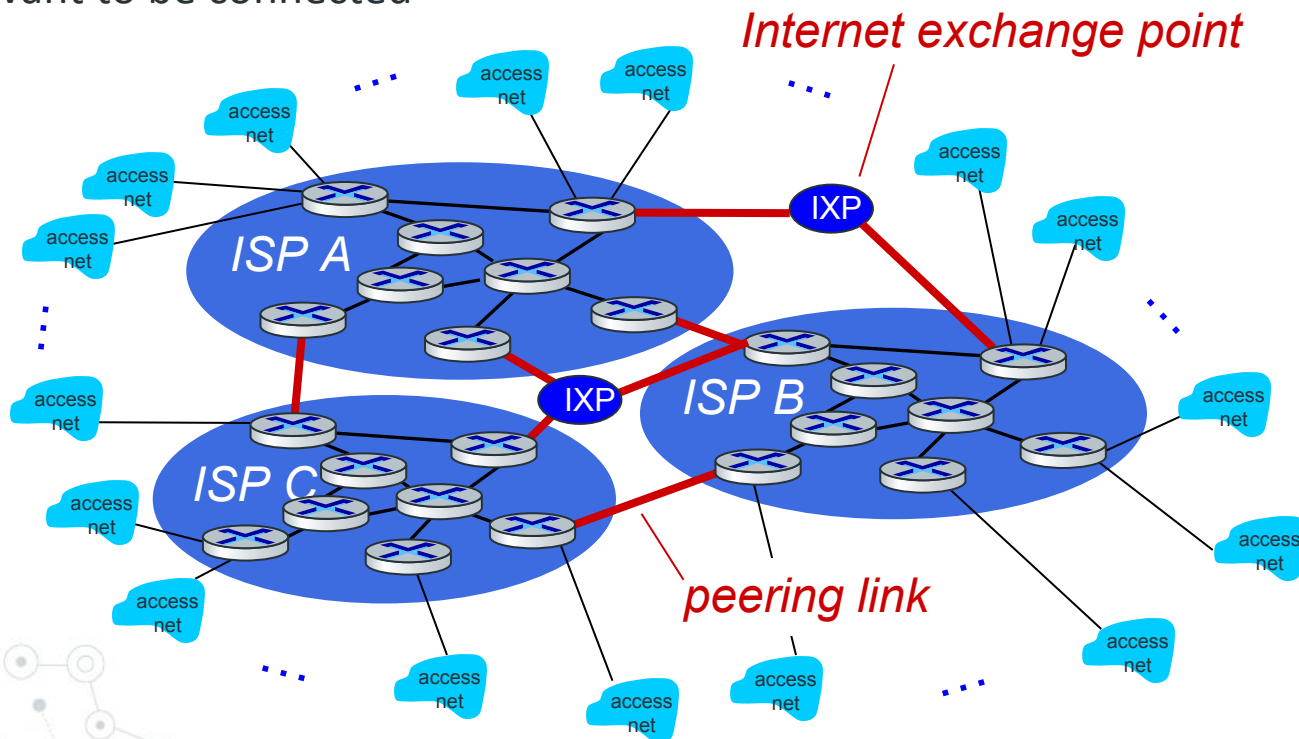
Internet structure: a “network of networks”

But if one global ISP is viable business, there will be competitors



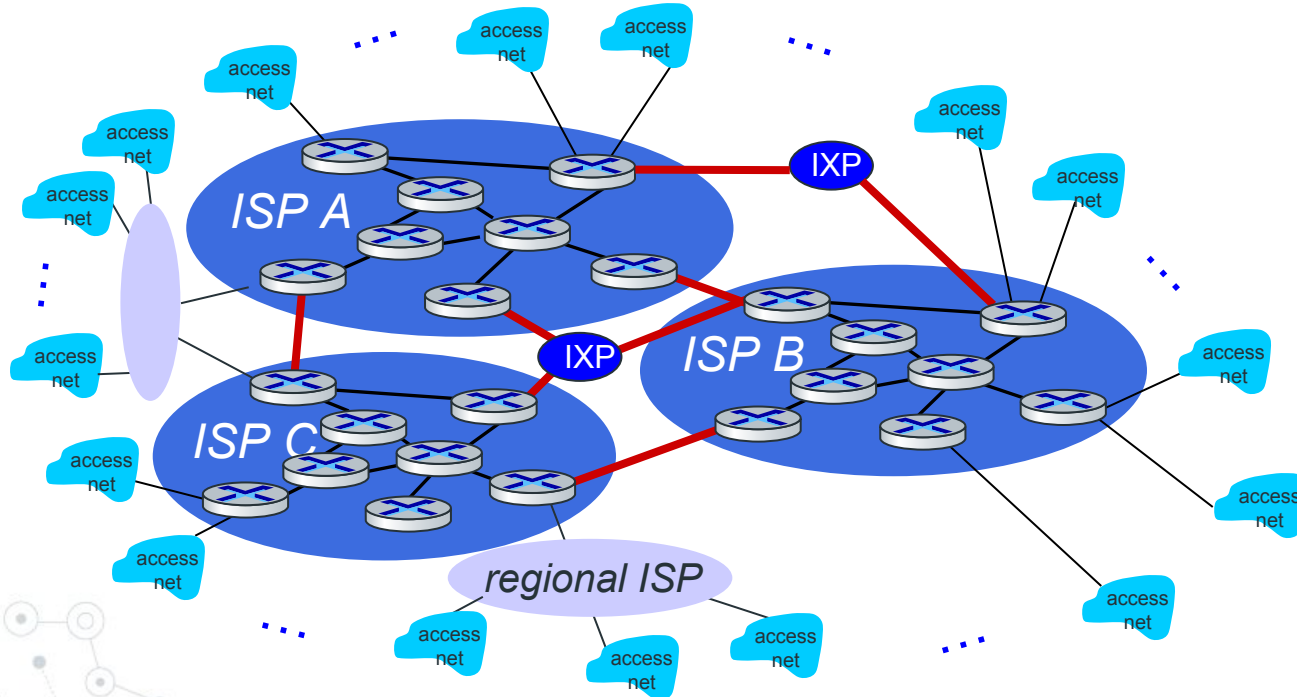
Internet structure: a “network of networks”

But if one global ISP is viable business, there will be competitors who will want to be connected



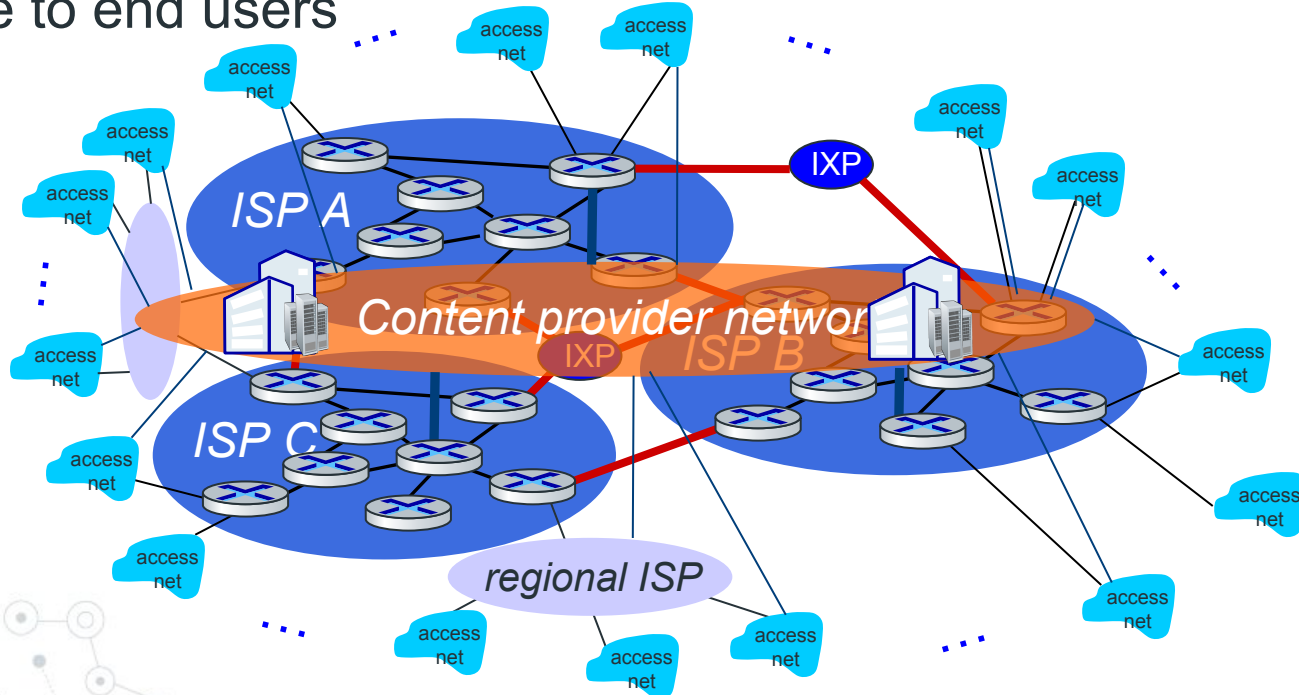
Internet structure: a “network of networks”

... and regional networks may arise to connect access nets to ISPs

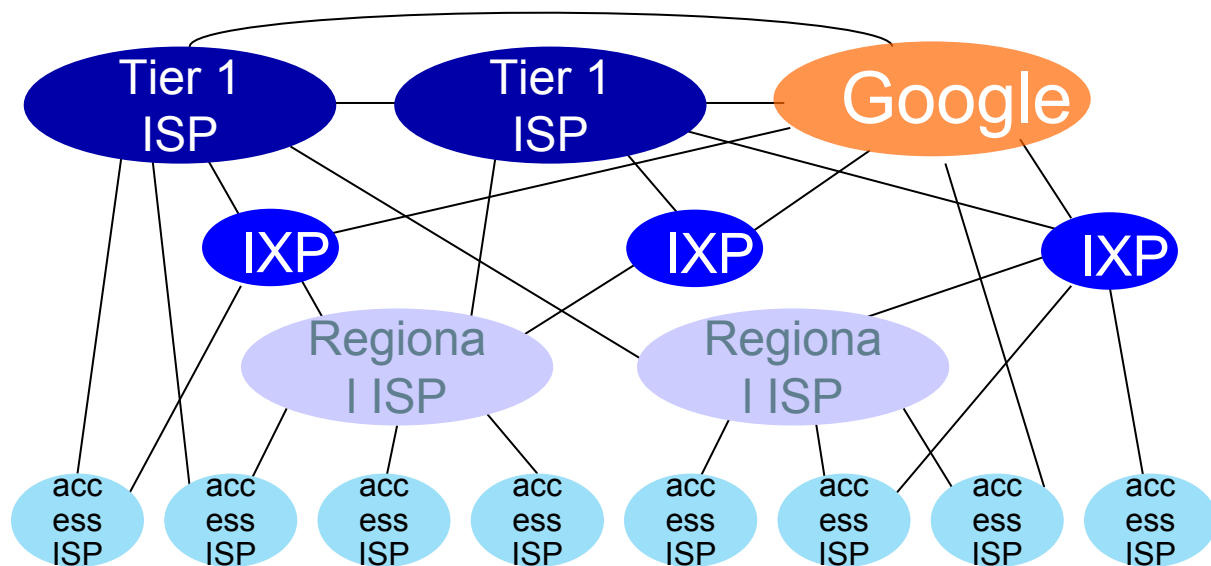


Internet structure: a “network of networks”

... and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet structure: a “network of networks”



At “center”: small # of well-connected large networks

- **“tier-1” commercial ISPs** (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- **content provider networks** (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

POP: point-of-presence
to/from other Sprint PoPS

links to peering networks

The diagram shows two network routers, represented as blue circles with a white 'X' inside. They are connected to each other by a thick black line. Each router has a thick black line extending from its top, labeled 'links to peering networks'. To the left of the first router, there is a vertical ellipsis and a horizontal line, indicating a connection to another network. Similarly, to the right of the second router, there is a vertical ellipsis and a horizontal line, indicating a connection to another network. The background of the slide features a network diagram with nodes and connections, and a yellow and blue curved shape on the left side.

